SIDE A · THEORY. The question says … → go to: "O, Ω, Θ", "which is faster", "true or false", "order these" → ⋈ **1** · "T(n) = …", "solve", "from first principles", "recursion tree" → ↻ **2** · "sorted array", "O(log n)", "O(log² n)", "divide and conquer", "design an algorithm" → ⌕ **3** · "sort", "stable", "pivot", "k-th smallest", "median", "which sort would you use" → ⇅ **4** · **Side B:** "expected", "on average", "random" → Σ **5** · "heap", "priority queue", "draw", "k-way", "largest k" → △ **6** · "lower bound", "at least … comparisons", "integers in range", "sort in O(n)" → ⊥ **7** · any formula → ✎ **8**. Boxes: ◉ meaning · § formal ▶ steps · ✎ answer shape ★ remember ⚠ pitfalls. Yellow-headed tables = patterns with worked examples.

⋈ 1 · COMPARING GROWTH (A / B / U)

◉ **IN WORDS** How fast do two running times grow for huge n? Constant factors and small n never matter. O = "grows no faster than" (ceiling), Ω = "at least as fast" (floor), Θ = "same rate". Faster algorithm = smaller running time. **A** = algorithm A is faster (T_A grows slower), **B** = B faster, **U** = cannot tell from the bounds given.

§ **FORMAL, write exactly** $f = O(g)$: $\exists c > 0$, n_0 with $f(n) \le c\,g(n)$ for all $n \ge n_0$. $f = \Omega(g)$: $f(n) \ge c\,g(n)$ for all $n \ge n_0$. $f = \Theta(g)$: both (prove each side). Worst case $\Theta(g)$: $O(g)$ on every input **and** some input family costs $\Omega(g)$.

▶ **HOW TO** ① Keep only the biggest term on each side; solve any recurrence or sum first (section 2). ② Find the matching row below. ③ Apply its trick. ④ T_A smaller ⇒ **A**; T_B smaller ⇒ **B**; same Θ ⇒ "same". Only O / Ω given ⇒ last row.

Shapes you see	Trick, then a worked example → answer
n^a vs n^b, maybe with logs	Compare exponents first; any n^ε beats any $(\log n)^k$. $n^{2.1}$ vs $n^2 \log^3 n$: $n^{0.1}$ beats $\log^3 n$ ⇒ **B**.
Ratios, products, close exponents	Take B/A: $\to\infty$ ⇒ **A** faster; $\to 0$ ⇒ **B**; constant ⇒ same. $n^{1.5}$ vs $n^2/\sqrt{\log n}$: $B/A = \sqrt{n/\log n} \to \infty$ ⇒ **A**.
Exponentials c^n, $a^{\log_b n}$	Compare bases: $2^{n/2} = (\sqrt{2})^n$; $4^n = 2^{2n} \ne \Theta(2^n)$; $2^{n+5} = \Theta(2^n)$; $a^{\log_b n} = n^{\log_b a}$. 5^n vs 7^n ⇒ **A**. $n\,2^n$ vs 3^n ⇒ **A** ($(3/2)^n$ beats n). n^2 vs $3^{\log_4 n} = n^{0.79}$ ⇒ **B**.
n inside an exponent, $(\log n)^{\log n}$, $n^{\log\log n}$	Take log of both sides (logs keep order) and compare those; or write both as $2^{h(n)}$. n^9 vs $2^{\sqrt{n}}$: $9\log n$ vs $\sqrt{n}$ ⇒ **A**. $(\log n)^{\log n}$ vs $n^{\log\log n}$: both logs $= \log n \cdot \log\log n$ ⇒ same.
$\sqrt{\log n}$, $\log\sqrt{n}$, $\log\log n$	Substitute $m = \log n$ (or $m = \sqrt{n}$) and compare simple functions of m. $\sqrt{\log n}$ vs $\log\sqrt{n} = \frac{1}{2}\log n$: $\sqrt{m}$ vs $m/2$ ⇒ **A**.
$n!$, n^n, $\log(n!)$	$\log(n!) = \Theta(n\log n)$; $(n/2)^{n/2} \le n! \le n^n$; $2^n = o(n!)$. 2^n vs $n!$ ⇒ **A**. $\log_3(n!)$ vs $n\log n$ ⇒ same.
Only O / Ω / Θ bounds given	Decided only if one side's **upper** bound is strictly below the other's **lower** bound. O vs O, Ω vs Ω, oscillating ($n^{\sin n}$) ⇒ **U**. $O(n^2)$ vs $\Omega(2^n)$ ⇒ **A**. $\Omega(n^3)$ vs $O(n^{2.81})$ ⇒ **B**. $O(n^2)$ vs $O(n^{2.3})$ ⇒ **U**. $O(\log n)$ vs $\Theta(\log\log n)$ ⇒ **U**.

Growth ladder, slow → fast

$1 < \log\log n < \log n < (\log n)^k < n^{0.01} < n < n\log n < n^2 < n^k < 2^n < n! < n^n$

- **Sums by hand.** $\sum_{i=1}^{n} i = \frac{n(n+1)}{2} = \Theta(n^2)$; $\sum i^c = \Theta(n^{c+1})$. Upper: n terms × biggest. Lower, **half trick**: keep the top $n/2$ terms, each $\ge (n/2)^c$.
- Geometric $\sum_{i\le k} r^i$: $r > 1 \Rightarrow \Theta(r^k)$ (last term) · $r < 1 \Rightarrow \Theta(1)$ (first term) · $r = 1 \Rightarrow k+1$. $H_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n} = \Theta(\log n)$ (blocks of $1, 2, 4, \ldots$ terms, each between $\frac{1}{2}$ and 1).

↻ 2 · RECURRENCES

◉ **IN WORDS** "Time for n = (number of calls) × (time for the smaller size) + (work in this call)". In $T(n) = a\,T(n/b) + f(n)$: a = calls, b = how much smaller, f = split + combine work. Mergesort: two sorts of $n/2$ plus n merging ⇒ $2T(n/2) + n$. Solving = unroll to the base case and add up what you collected; the recursion tree is the same sum drawn row by row (★ below).

▶ **HOW TO** "From first principles": ① draw the tree, write row i cost and the leaves (★ below); ② put $L = \log_b n$, do the sum (section 8); ③ write "leaves give …, rows sum to …, so $T(n) = \Theta(\ldots)$" and say what wins: leaves, equal rows, or root.

★ **TOTAL COST of $T(n) = aT(n/b) + f(n)$, $T(1) = 1$. $L = \log_b n$ levels**

Row i has a^i nodes of size n/b^i ⇒ row cost $a^i f(n/b^i)$. Leaves: $a^L = n^{\log_b a}$, cost 1 each. $T(n) = n^{\log_b a} + \sum_{i=0}^{L-1} a^i f(n/b^i)$ = leaves + sum of rows. Sum via section 8 (geometric, or $1 + \cdots + L = L(L+1)/2$).

Worked: $T(n) = 4T(n/4) + n\log_4 n$

Row i: $4^i \cdot \frac{n}{4^i}\log_4 \frac{n}{4^i} = n(L-i)$ (4^i cancels; $\log(a/b) = \log a - \log b$). Leaves $4^L = n$. $T(n) = n + n(L + (L-1) + \cdots + 1) = n + n\frac{L(L+1)}{2} = \Theta(n\log^2 n)$.

★ **REMEMBER Which case am I in?** Write $f(n) = n^c \log^k n$ and $d = \log_b a$ (the leaves exponent, so leaves cost n^d). Two tricks that simplify row i: $a^i \cdot \frac{n^c}{b^{ic}} = n^c \left(\frac{a}{b^c}\right)^i$ and $\log_b \frac{n}{b^i} = \log_b n - i$. Compare c with d:

Case	Rows do	Answer $\Theta(n^\alpha \log^\beta n)$	α	β
$c < d$	grow toward leaves	$\Theta(n^d)$, leaves win	d	0
$c = d$	all about equal	$\Theta(n^d \log^{k+1} n)$	d	$k+1$
$c > d$	shrink from root	$\Theta(n^c \log^k n)$, root wins	c	k

Recurrence	Θ	Recurrence	Θ
$T(n/2) + 1$	$\log n$	$T(n/2) + n$	n
$2T(n/2) + n$	$n\log n$	$2T(n/2) + 1$	n
$2T(n/2) + n\log n$	$n\log^2 n$	$2T(n/2) + n^2$	n^2
$T(n-1) + n$	n^2	$T(n/2) + \log n$	$\log^2 n$
$T(\sqrt{n}) + 1$	$\log\log n$	$T(\sqrt{n}) + \log n$	$\log n$
$T(\alpha n) + T(\beta n) + n$, $\alpha + \beta = 1$	$n\log n$	same, $\alpha + \beta < 1$	n

- $T(\sqrt{n})$: set $m = \log n$ ($n = 2^m$, $\sqrt{n} = 2^{m/2}$) ⇒ $S(m) = S(m/2) + f$, solve, substitute back.
- $T(n-1) + f(n)$ is just $\sum_{i\le n} f(i)$: $T(n-1) + \log n = \log(n!) = \Theta(n\log n)$. $2T(n-1) + 1$ doubles each step: $2^n - 1$ (Hanoi).

⌕ 3 · BINARY SEARCH & DIVIDE AND CONQUER

◉ **IN WORDS** Split the input into smaller copies of the same problem, solve them recursively, combine. The required running time tells you the shape:

Target time	Shape to use
$O(\log n)$ on a sorted or monotone array	one binary search: halve, $O(1)$ work per step
$O(\log^2 n)$	$\log n$ binary searches, or a binary search inside a binary search
$O(\log k)$, $k \ll n$	gallop: double an index until you pass k, then binary search that range
$O(n\log n)$	two halves + $O(n)$ combine (mergesort shape), or sort then one pass
$O(n\log k)$	heap of size k (section 6), or recursion that stops at depth $\log k$
$O(n)$ expected / $O(n)$ with "integers in a range"	random pivot, recurse on one side (section 4) / counting or radix sort (section 7)

```
LowerBound(A[1..n], x) → first i with A[i] ≥ x (n+1 if none).
"<" → "≤" gives first A[i] > x
  l ← 1, r ← n+1;  while l < r: m ← ⌊(l+r)/2⌋; if A[m] < x
then l ← m+1 else r ← m;  return l
Rotated sorted array, index of max: m ← middle; if A[m] >
A[m+1] return m;
  if A[m] ≥ A[l] then recurse on the right half else on the
left half                O(log n)
Answer k is small: i ← 1; while A[i] = 0: i ← 2i; then binary
search in A[i/2..i]  O(log k)
```

Problem	Idea in one line	Time
Mergesort; count inversions ($i < j$, $A[i] > A[j]$)	sort both halves, Merge with an ∞ sentinel; for inversions, when an item of the right half is copied out add the number of left-half items still waiting	$2T(n/2) + n = \Theta(n\log n)$
Max subarray (best buy/sell pair = same on daily differences)	best in left, best in right, best crossing the middle = best suffix of left + best prefix of right, two linear scans	$2T(n/2) + n = \Theta(n\log n)$; Kadane $O(n)$
Multiply n-bit integers	halves: 4 products ⇒ $\Theta(n^2)$; Karatsuba: middle term $= (a_1+a_0)(b_1+b_0) - a_1b_1 - a_0b_0$ ⇒ 3 products. Matrices: Strassen 7 block products ⇒ $n^{2.807}$	$3T(n/2) + n = \Theta(n^{1.585})$
k-th smallest of two sorted arrays	compare $A[k/2]$ with $B[k/2]$; the smaller side's first $k/2$ cannot hold the answer, drop them, $k \leftarrow k/2$; $k = 1$: min of heads	$\Theta(\log k)$
Distinct values of a sorted array, d of them	from the current value, LowerBound its successor, repeat	$O(d\log n)$

✎ **ANSWER SHAPE** Design answer: ① idea in two sentences; ② "Procedure Name(input) → output"; ③ steps or pseudocode; ④ name each black box with its cost ("Select, $\Theta(n)$"); ⑤ recurrence or sum, solved, final Θ; ⑥ one line why correct.

⇅ 4 · SORTING & SELECTION

```
Partition(A, l, r): pivot x ← A[r]; i ← l-1
  for j ← l to r-1: if A[j] ≤ x then i ← i+1, swap A[i] ↔ A[j]
  swap A[i+1] ↔ A[r]; return i+1                  Θ(n), in
place; A[l..i] ≤ x, A[i+1..j-1] > x
Quicksort(A, l, r): if l ≥ r return; m ← Partition;
Quicksort(l, m-1); Quicksort(m+1, r)
Select(A, l, r, k) → k-th smallest: random pivot swapped to
A[r]; m ← Partition; rank ← m-l+1
  if k = rank return A[m]; if k < rank: Select(l, m-1, k) else
Select(m+1, r, k - rank)
```

Claim	Proof in words (write these lines)
Random quicksort: $\Theta(n\log n)$ expected comparisons	Sorted values $z_1 < \cdots < z_n$. z_i, z_j are compared iff the first pivot chosen from $\{z_i, \ldots, z_j\}$ is z_i or z_j (a pivot between separates them). Uniform over $j - i + 1$ values ⇒ $\Pr = 2/(j-i+1)$. $E = \sum_{i<j} 2/(j-i+1)$; each i contributes $\le 2H_n = O(\log n)$; n values of i ⇒ $O(n\log n)$.
Random Select: $O(n)$ expected	A pivot is *good* if it lands in the middle half: $\Pr = \frac{1}{2}$, and it throws away $\ge \frac{1}{4}$ of the array. After i good pivots the size is $\le (\frac{3}{4})^i n$; 2 calls per good pivot on average ⇒ $E \le \sum_i 2(\frac{3}{4})^i n = 8n$.

Sort	Worst	Stable	Space	Pick it when
Insertion	n^2, best n	yes	$O(1)$	"almost sorted", few inversions: $O(n+I)$
Mergesort	$n\log n$	yes	$O(n)$	"stable" (equal keys keep order), linked lists
Quicksort	n^2 , exp. $n\log n$	no	$O(\log n)$	"in practice"; worst: sorted input, last pivot
Heapsort	$n\log n$	no	$O(1)$	"in place", guaranteed $n\log n$
Counting / Radix	$n+k$ / $d(n+k)$	yes	$O(n+k)$	"ints in $[0,k]$" / "below n^c", c passes

SIDE B · APPLICATIONS. The question says … → go to: "expected number of …", "on average", "how many … until", "probability" → **Σ 5** · "min-heap", "max-heap", "priority queue", "draw the heap after", "build-heap", "k sorted lists", "largest k", "running median", "sliding window" → **△ 6** · "any algorithm needs at least", "lower bound", "decision tree", "integers in the range", "sort in O(n)", "stable", "radix" → **⊥ 7** · logs, sums, Stirling, binomials, e limits → **✎ 8**. Side A: ⋈ 1 compare growth · ↻ 2 recurrences · ⌕ 3 search and divide and conquer · ⇅ 4 sorting and selection.

Σ 5 · EXPECTATION & RANDOMIZED ALGORITHMS

◉ IN WORDS Expected value = long-run average. To find "how many X happen on average", never compute it directly: split the count into yes/no questions ("does person i get their own hat?"); the average of a yes/no question is just its probability; the average of a sum is the sum of the averages, always, even when the questions depend on each other. Hats: n people × $\frac{1}{n}$ = 1.

§ FORMAL, write exactly Indicator: $X_i = 1$ if event i happens, else 0; $E[X_i] = \Pr[X_i = 1]$. Linearity: $E[X+Y] = E[X] + E[Y]$ for any random variables, independent or not; so $E[\sum X_i] = \sum \Pr[\text{event } i]$. Waiting time: success probability p per try ⇒ expected $1/p$ tries. Tail sum: $E[X] = \sum_{i \geq 1} \Pr[X \geq i]$. Randomized running time = expected time on the worst input.

Question says	Pattern	Worked example
"expected number of [things] that …"	one indicator per thing; add the probabilities	Own hat: $\Pr = \frac{1}{n}$ each ⇒ $E = 1$. Own hat or a better one: $\Pr = \frac{1}{n}$ ⇒ $E = \frac{n+1}{2}$.
"expected number of pairs that …"	one indicator per pair; $\binom{n}{2}$ pairs	Birthdays, k people, n days: $E = \binom{k}{2}\frac{1}{n} = \frac{k(k-1)}{2n}$; reaches 1 at $k \approx \sqrt{2n}$.
"how many boxes / tries until you have …"	stages; stage i succeeds with p_i per step; add $1/p_i$	Coupon collector, n types: with i owned, $p = \frac{n-i}{n}$ ⇒ $E = \sum_{i=0}^{n-1} \frac{n}{n-i} = nH_n = \Theta(n \log n)$. First $n/2$ types: every $p \geq \frac{1}{2}$ ⇒ $E \leq n$.
"expected number of distinct … after m draws"	indicator per type; $\Pr[\text{seen}] = 1 - (1 - \frac{1}{n})^m$	n draws, n types: $E = n\big(1 - (1 - \frac{1}{n})^n\big) \approx n(1 - \frac{1}{e}) \approx 0.63n$.
"best so far", "record", "hire", "new maximum"	$\Pr[i\text{-th is the best of the first } i] = \frac{1}{i}$	$E = \sum_{i=1}^{n} \frac{1}{i} = H_n = \Theta(\log n)$. Cards guessed with memory: $\frac{1}{n-i+1}$, also H_n. Hash collisions: $\frac{1}{m}$ each ⇒ $\frac{n}{m}$.
"expected comparisons / calls of a randomized algorithm"	indicator per pair compared (quicksort), or good / bad stages (Select)	Quicksort: $\Pr[z_i, z_j \text{ compared}] = \frac{2}{j-i+1}$ ⇒ $\Theta(n \log n)$. Select, random-probe search: a good step ($\Pr = \frac{1}{2}$) removes $\frac{1}{4}$ ⇒ $\sum 2(\frac{3}{4})^i n = O(n)$.
"repeat until it works"	success p per round ⇒ $1/p$ rounds × cost per round	Majority by try-and-check: $p > \frac{1}{2}$ ⇒ ≤ 2 rounds of $O(n)$.

✎ ANSWER SHAPE "Let $X_i = 1$ if [event i], else 0, for $i = 1..n$. $\Pr[X_i = 1] = \dots$ because [reason]. $X = \sum X_i$ is [the quantity]. By linearity of expectation $E[X] = \sum_i \Pr[X_i = 1] = \cdots = \Theta(\dots)$." Waiting times: "Stage i ends with probability p_i per step, so it lasts $1/p_i$ steps on average; total $\sum 1/p_i$."

⚠ PITFALLS Excluding yourself: $(1 - \frac{1}{n})^{n-1}$, not $(1 - \frac{1}{n})^n$ · never assume independence, sums do not need it · write the exact form when asked, then simplify with $H_n \approx \ln n$, $(1 - \frac{1}{n})^n \to \frac{1}{e}$.

△ 6 · HEAPS

◉ IN WORDS A min-heap is a binary tree filled level by level from the left, where every parent ≤ its children, so the smallest item is always on top (find-min $O(1)$); siblings are in no order. Delete or change a key: bubble up if smaller, sift down if larger, $O(\log n)$. It lives in a plain array: node i has children $2i$, $2i+1$ and parent $\lfloor i/2 \rfloor$; height $\lfloor \log_2 n \rfloor$. Max-heap: same with ≥.

Operation	Steps	Cost
Insert x	put x in the next free slot $H[n+1]$; while $x <$ parent, swap up	$O(\log n)$
Extract-min	take $H[1]$; move the last item $H[n]$ to $H[1]$, $n \leftarrow n-1$; while it is $>$ a child, swap with the **smaller** child	$O(\log n)$
Build-heap	sift down nodes $\lfloor n/2 \rfloor, \dots, 1$ in that order (leaves need nothing)	$O(n)$
Heapsort	build a max-heap; n times: swap $H[1] \leftrightarrow H[n]$, $n \leftarrow n-1$, sift down $H[1]$	$O(n \log n)$, in place, not stable

▶ HOW TO Drawing "insert these one by one" (0.5 off per mistake): for **every** insert ① write the array with the new number appended, ② compare with its parent at $\lfloor i/2 \rfloor$, swap while smaller, writing the array after each swap, ③ draw the tree. Check: bottom level filled from the left, every parent ≤ its children.

Insert 4, 3, 8, 2 into an empty min-heap, then extract-min

4: $[4]$ · 3: $[4,3]$, $3 < 4$ swap ⇒ $[3,4]$ · 8: $[3,4,8]$, $8 \geq 3$ stays · 2: $[3,4,8,2]$, parent of slot 4 is slot 2 (holds 4): swap ⇒ $[3,2,8,4]$; parent of slot 2 is slot 1 (holds 3): swap ⇒ $[2,3,8,4]$. Tree: 2 on top, children 3 and 8, 4 under 3. Extract-min: take 2; last item 4 goes on top ⇒ $[4,3,8]$; smaller child 3 ⇒ swap ⇒ $[3,4,8]$.

★ REMEMBER Build-heap is $O(n)$: a node at height h costs $O(h)$ and at most $n/2^{h+1}$ nodes have height h, so total $\leq \sum_h h \cdot \frac{n}{2^{h+1}} = \frac{n}{2} \sum_h \frac{h}{2^h} = \frac{n}{2} \cdot 2 = O(n)$. Most nodes are near the bottom and barely move. Inserting one by one costs $\sum \log i = \Theta(n \log n)$.

Question says	Design	Time
"merge k sorted lists", N items	min-heap of k entries (value, list, position); pop the min, output, push the next item of that list	$O(N \log k)$, space $O(k)$
"each element at most k from its sorted place"	min-heap of the next $k+1$ items; pop, push the next	$O(n \log k)$
"largest (smallest) k of n"	min-heap of size k: push each item, pop the min when size exceeds k. Or Select the k-th in $O(n)$ then sort k items	$O(n \log k)$; $O(n + k \log k)$
"median after every insertion"	max-heap of the lower half, min-heap of the upper half, sizes differ ≤ 1; insert on the right side, move one root if unbalanced	insert $O(\log n)$, median $O(1)$
"maximum of every window of width W"	max-heap of (value, index); pop roots whose index left the window (lazy; heap may grow to n); with a position map delete the leaver at once, heap stays size W	$O(n \log n)$ lazy; $O(n \log W)$ with map
"$O(n \log \log n)$", "largest $\log n$ items"	$n/\log n$ blocks of $\log n$, sort each ($\log n \cdot \log \log n$ each); heap of block tops; extract-max $\log n$ times, pushing the next from that block	$n \log \log n + n + \log^2 n$
"m sets, global min $O(1)$, insert $O(\log m + \log n_i)$"	one heap per set + a top heap of the m minima; each set keeps a pointer to its slot in the top heap; a new x below its set's min overwrites that slot and bubbles up, fixing pointers on every swap	as required

⊥ 7 · LOWER BOUNDS & LINEAR-TIME SORTING

◉ IN WORDS Each comparison is a yes/no question; q questions can tell apart at most 2^q cases. If a problem has N possible answers you need $2^q \geq N$, so $q \geq \log_2 N$. Sorting has $n!$ answers and $\log(n!) \approx n \log n$, so no comparison sort beats $n \log n$. Counting and radix sort never compare keys, so they escape, for small integer keys only.

▶ HOW TO Lower bound, four lines: ① count the distinct possible outputs N; ② "any comparison-based algorithm is a binary decision tree with at least N leaves"; ③ "a binary tree of height h has at most 2^h leaves, so $h \geq \log_2 N$"; ④ simplify; height = worst-case comparisons.

Question says	Outputs N	Bound
"sort n items"	$n!$	$\log_2 n! \geq \frac{n}{2} \log_2 \frac{n}{2} = \Omega(n \log n)$
"the k smallest in sorted order"	$\frac{n!}{(n-k)!} \geq (n/2)^k$	$\Omega(k \log n)$
"merge two sorted arrays of n"	$\binom{2n}{n} \approx 4^n/\sqrt{\pi n}$	$\approx 2n = \Omega(n)$
"split into k sorted blocks of n/k"	$n!/((n/k)!)^k$	$\Omega(n \log k)$

```
CountingSort(A[1..n], keys 0..k): C[v] ← number of j with A[j] = v
  for v ← 1 to k: C[v] ← C[v] + C[v-1]                now C[v] = number of keys ≤ v = last slot for v
  for j ← n down to 1: B[C[A[j]]] ← A[j]; C[A[j]] ← C[A[j]] - 1
                                  Θ(n + k), stable
RadixSort: for each digit, least significant first:
CountingSort on that digit (must be stable)
```

Question says	What to do	Why / time
"integers in $[0,k]$", $k = O(n)$	counting sort	$\Theta(n+k)$
"integers below n^c" (e.g. n^3)	view each key as c digits in base n; radix sort, c passes	$c \times \Theta(n+n) = \Theta(n)$. Linear ⇔ range polynomial ⇔ keys have $O(\log n)$ bits. Range $n^{\log n}$ ⇒ $\log n$ passes ⇒ $\Theta(n \log n)$, not linear
keys are not integers (fractions p/q, $p, q \leq n$)	build an integer key that keeps the order: distinct fractions differ by $\geq 1/n^2$, so $K = \lfloor n^2 p/q \rfloor \in [n, n^3]$ preserves order; radix sort K	$\Theta(n)$. No gcd ($\log n$ each), no floats (rounding merges close values)
"why scan backwards", "why must each pass be stable"	backwards scan hands equal keys their slots right to left, so input order survives; radix is correct by induction on digits, and that step needs stability	forward scan reverses ties

✎ 8 · FORMULAS

Logs	Exponents	Sums	Approximations
$\log(ab) = \log a + \log b$ $\log(a^b) = b \log a$ $\log_b a = \frac{\log a}{\log b} = \frac{1}{\log_a b}$ $\log_c n = \Theta(\log_2 n)$ $\log(n!) = \Theta(n \log n)$ $\log_2 3 \approx 1.585$, $\log_2 7 \approx 2.807$, $\log_4 3 \approx 0.79$	$a^{\log_b n} = n^{\log_b a}$ $2^{\log_2 n} = n$ $(c^m)^n = c^{mn}$, $c^m c^n = c^{m+n}$ $4^n = 2^{2n} \neq \Theta(2^n)$ $2^{n+c} = \Theta(2^n)$ $n^{1/\log n} = 2$	$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$ $\sum_{i=0}^{k} r^i = \frac{r^{k+1}-1}{r-1}$ $\sum_{i \geq 0} x^i = \frac{1}{1-x}$, $x < 1$ $\sum_{i \geq 1} i x^i = \frac{x}{(1-x)^2}$; $\sum \frac{i}{2^i} = 2$ $H_n = \sum_{i=1}^{n} \frac{1}{i} \approx \ln n + 0.58$ $\sum_{i \leq n} \log i = \log(n!)$	$n! \approx \sqrt{2\pi n}\,(n/e)^n$ $(n/2)^{n/2} \leq n! \leq n^n$ $\binom{n}{k} \leq \frac{n^k}{k!} \leq (\frac{en}{k})^k$ $\binom{2n}{n} \approx \frac{4^n}{\sqrt{\pi n}}$ $(1 - \frac{1}{n})^n \to \frac{1}{e} \approx 0.37$, $(1 + \frac{1}{n})^n \to e$ $1 - x \leq e^{-x}$, so $(1-p)^k \leq e^{-pk}$